

Partial Algebra

1) Fie $P_1(x) = 1 + 3x - 5x^2$; $P_2(x) = 7 + 42x - 53x^2$; $P_3(x) = -7 - 56x + 65x^2 \in \mathbb{R}_2[x]$

și subspațiul:

$$W = \{ P \in \mathbb{R}_2[x] \mid P(0) + 3P'(0) - 5P''(0) = 0, -7P'(0) + 6P''(0) = 0 \}$$

a) Arătați că P_1, P_2, P_3 = linear dependente;

Fie $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$, a.î.: $\alpha_1 P_1 + \alpha_2 P_2 + \alpha_3 P_3 = 0 (=)$

$$\Leftrightarrow \begin{cases} \alpha_1 + 7\alpha_2 - 7\alpha_3 = 0 \\ 3\alpha_1 + 42\alpha_2 - 56\alpha_3 = 0 \\ -5\alpha_1 - 53\alpha_2 + 65\alpha_3 = 0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 7 & -7 & 0 \\ 3 & 42 & -56 & 0 \\ 5 & -53 & 65 & 0 \end{array} \right) \xrightarrow[\substack{L_2 - 3L_1 \\ L_3 + 5L_1}]{L_2 - 3L_1} \left(\begin{array}{ccc|c} 1 & 7 & -7 & 0 \\ 0 & 21 & -77 & 0 \\ 0 & 167 & -167 & 0 \end{array} \right)$$

cu $\alpha_3 = -3$ (din ipoteză) $\Rightarrow \begin{cases} \alpha_1 + 7\alpha_2 = -21 \\ 3\alpha_1 + 42\alpha_2 = -168 \\ -5\alpha_1 - 53\alpha_2 = 195 \end{cases}$

$$\begin{cases} \alpha_1 + 7\alpha_2 = -21 \\ 3\alpha_1 + 42\alpha_2 = -168 \end{cases} \xrightarrow{(-3)} \begin{cases} -3\alpha_1 - 21\alpha_2 = 63 \\ 3\alpha_1 + 42\alpha_2 = -168 \end{cases}$$

$$21\alpha_2 = -105 \Rightarrow \boxed{\alpha_2 = -5}$$

$$\alpha_1 - 35 = -21 \Rightarrow \boxed{\alpha_1 = 14}$$

cu $\alpha_1, \alpha_2, \alpha_3 \neq 0 \Rightarrow P_1, P_2, P_3$ = lin. dependente = 1

$$\Rightarrow \alpha_1 P_1 + \alpha_2 P_2 + \alpha_3 P_3 = 0 \Leftrightarrow \boxed{14P_1 - 5P_2 - 3P_3 = 0}$$

e) Analizăm fiecare relație din W . Pentru asta considerăm un polinom $P \in \mathbb{R}_2[X]$, $P(x) = a_0 + a_1x + a_2x^2$, $a_0, a_1, a_2 \in \mathbb{R}$

$$P'(x) = a_1 + 2a_2x \Rightarrow P'(0) = a_1$$

$$P''(x) = 2a_2 \Rightarrow P''(0) = 2a_2$$

$$P(0) = a_0$$

$$P(0) + P'(0) - 5P''(0) = 0 \Leftrightarrow a_0 + a_1 - 10a_2 = 0 \Rightarrow a_0 = -a_1 + 10a_2 = \frac{7}{10}a_2 - \frac{36}{7}a_2 = \frac{34}{7}a_2$$

$$-7P'(0) + 6P''(0) = 0 \Leftrightarrow -7a_1 + 12a_2 = 0 \Rightarrow a_1 = \frac{12}{7}a_2$$

Deci: $P(x) = a_0 + a_1x + a_2x^2 = \frac{34}{7}a_2 + \frac{12}{7}a_2x + a_2x^2 =$

$$= \frac{a_2}{7} (34 + 12x + 7x^2) \Rightarrow W = \text{Sp} \{34 + 12x + 7x^2\} \Rightarrow$$

$$\Rightarrow \boxed{\dim_{\mathbb{R}_2[X]} W = 1}; \text{ Notăm } P = 34 + 12x + 7x^2$$

c) ~~deoarece~~ P_1, P_2, P_3 = lin. dependente și $14P_1 - 5P_2 - 3P_3 = 0 \Rightarrow$

$\Rightarrow P_1 = \frac{5}{14}P_2 + \frac{3}{14}P_3 \Rightarrow B = \text{Sp} \{P_1, P_2, P_3\} = \text{Sp} \{P_2, P_3\}$, care este o bază deoarece P_2, P_3 = lin. independente.

Deci $W + B = \text{Sp} \{P_2, P_3, P\} = \mathbb{R}_2[X] \Leftrightarrow P_2, P_3, P$ = lin. independenți.

Studiem liniar independența lui P_2, P_3, P : Fie $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ a.t. $\lambda_1P_2 + \lambda_2P_3 + \lambda_3P = 0$ ($\Rightarrow$)

$$\Leftrightarrow \begin{cases} 7\lambda_1 - 7\lambda_2 + 34\lambda_3 = 0 \\ 42\lambda_1 - 56\lambda_2 + 12\lambda_3 = 0 \\ -53\lambda_1 + 65\lambda_2 + 7\lambda_3 = 0 \end{cases}$$

Luăm coordonatele fiecărui polinom în parte și le scriem în baza canonică:

$$[P_2]_{B_c} = \begin{pmatrix} 7 \\ 42 \\ -53 \end{pmatrix}$$

$$[P_3]_{B_c} = \begin{pmatrix} -7 \\ -56 \\ 65 \end{pmatrix}$$

$$[P]_{B_c} = \begin{pmatrix} 34 \\ 12 \\ 7 \end{pmatrix}$$

$$\text{Calculăm: } \begin{vmatrix} 7 & -7 & 34 \\ 42 & -56 & 12 \\ -53 & 65 & 7 \end{vmatrix} = -49 \cdot 56 + 42 \cdot 65 \cdot 34 - \begin{vmatrix} -4 & 2 & 53 \\ -11 & -9 & 19 \\ -53 & 65 & 7 \end{vmatrix} =$$

$$= 36 \cdot 7 - 11 \cdot 65 \cdot 53 - 53 \cdot 2 \cdot 19 - 53^2 \cdot 9 + 19 \cdot 65 \cdot 4 + 7 \cdot 2 \cdot 11 =$$

$$= -59844 \neq 0 \Rightarrow P_2, P_3, P = \text{liniar independenți} \Rightarrow$$

$$\Rightarrow W + B = \mathbb{R}_2[X], \text{ deci } W \text{ este un complement al lui } B = \text{Sp}\{P_2, P_3\}$$

2) a) Se consideră $A = \begin{pmatrix} -6 & -1 \\ -8 & -4 \end{pmatrix}$ și aplicația liniară:

$$f: \mathcal{M}_2(\mathbb{R}) \rightarrow \mathcal{M}_2(\mathbb{R}), f(X) = AX - XA$$

$$a) \text{ Ker}(f) = \{ \cancel{B} \in \mathcal{M}_2(\mathbb{R}) \mid f(B) = O_2 \}$$

$$B = \begin{pmatrix} a & c \\ c & d \end{pmatrix} \Rightarrow f(B) = AB - BA = O_2 \Rightarrow AB = BA \Rightarrow$$

$$\Leftrightarrow \begin{pmatrix} -6 & -1 \\ -8 & -4 \end{pmatrix} \cdot \begin{pmatrix} a & c \\ c & d \end{pmatrix} = \begin{pmatrix} a & c \\ c & d \end{pmatrix} \cdot \begin{pmatrix} -6 & -1 \\ -8 & -4 \end{pmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{pmatrix} -6a - c & -6c - d \\ -8a - 4c & -8c - 4d \end{pmatrix} = \begin{pmatrix} -6a - 8c & -a - 4c \\ -6c - 8d & -c - 4d \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} 6a + c = 6a + 8c \Rightarrow \boxed{c = 8c} \\ 8c + 4d = c + 4d \\ 6c + 8d = 8a + 4c \Rightarrow 8(a - d) = 2c \Rightarrow 8(a - d) = 16c \Rightarrow a - d = 2c \\ 6c + d = a + 4c \end{cases}$$

$$B_{\mathcal{C}} = \begin{pmatrix} 2e_1 + d & e_1 \\ 8e_1 & d \end{pmatrix}$$

$$B = \begin{pmatrix} 2e_1 + d & e_1 \\ 8e_1 & d \end{pmatrix} = \begin{pmatrix} 2e_1 & e_1 \\ 8e_1 & 0 \end{pmatrix} + \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix} =$$

$$= e_1 \begin{pmatrix} 2 & 1 \\ 8 & 0 \end{pmatrix} + d \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \text{Ker } f = \text{Sp} \left\{ \begin{pmatrix} 2 & 1 \\ 8 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$$\Rightarrow \dim_{\mathbb{R}} \text{Ker}(f) = 2$$

$$\text{Im}(f) = \left\{ D \in M_2(\mathbb{R}) \mid \exists C \in M_2(\mathbb{R}) \text{ p.n. } f(C) = D \right\}$$

$$C = \begin{pmatrix} x & y \\ z & t \end{pmatrix}, D = \begin{pmatrix} a & e_1 \\ c & d \end{pmatrix}$$

$$f(C) = D \Leftrightarrow AC - CA = D \Leftrightarrow (-1) \cdot \begin{pmatrix} 6 & 1 \\ 8 & 4 \end{pmatrix} \begin{pmatrix} x & y \\ z & t \end{pmatrix} + \begin{pmatrix} x & y \\ z & t \end{pmatrix} \begin{pmatrix} 6 & 1 \\ 8 & 4 \end{pmatrix} =$$

$$= \begin{pmatrix} a & e_1 \\ c & d \end{pmatrix} \Leftrightarrow \begin{pmatrix} 6x+z & 6y+t \\ 8x+4z & 8y+4t \end{pmatrix} + \begin{pmatrix} 6x+8y & x+4y \\ 6z+8t & z+4t \end{pmatrix} = \begin{pmatrix} a & e_1 \\ c & d \end{pmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{pmatrix} 8y-z & x-2y-t \\ -8x+2z+8t & z-8y \end{pmatrix} = \begin{pmatrix} a & e_1 \\ c & d \end{pmatrix} \Rightarrow$$

$$\Leftrightarrow \begin{cases} 8y-z = a \\ x-2y-t = e_1 \\ -8x+2z+8t = c \\ z-8y = d \end{cases} \Rightarrow \begin{cases} a+d=0 \text{ (prin adunarea relatiei 1 cu relatia 4)} \\ d = -a \end{cases}$$

$$\left(\begin{array}{cccc|c} 0 & 8 & -1 & 0 & a \\ 1 & -2 & 0 & -1 & e_1 \\ -8 & 0 & 2 & 8 & c \\ 0 & 0 & 1 & -8 & d \end{array} \right) \xrightarrow{L_2 \leftrightarrow L_1} \left(\begin{array}{cccc|c} 1 & -2 & 0 & -1 & e_1 \\ 0 & 8 & -1 & 0 & a \\ -8 & 0 & 2 & 8 & c \\ 0 & 0 & 1 & -8 & d \end{array} \right) \xrightarrow{L_3 + 8L_1}$$

$$\rightarrow \left(\begin{array}{cccc|c} 1 & -2 & 0 & -1 & e_1 \\ 0 & 8 & -1 & 0 & a \\ 0 & -16 & 2 & 0 & c+8a \\ 0 & 0 & 1 & -8 & d \end{array} \right) \xrightarrow{L_3 + 2L_2} \left(\begin{array}{cccc|c} 1 & -2 & 0 & -1 & e_1 \\ 0 & 8 & -1 & 0 & a \\ 0 & 0 & 0 & 0 & c+10a \\ 0 & 0 & 1 & -8 & d \end{array} \right)$$

Answer:

$$\begin{cases} c+10a=0 \Rightarrow c=-10a \\ 8y-z=a \\ x-2y-z=e_1 \\ z-8t=d \end{cases}$$

$$\text{Def: } D = \begin{pmatrix} a & e_1 \\ -10a & -a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ -10 & -1 \end{pmatrix} + e_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} =$$

$$\Rightarrow \text{Im}(f) = \text{Sp} \left\{ \begin{pmatrix} 1 & 0 \\ -10 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\} \Rightarrow \boxed{\dim_{\mathbb{R}} \text{Im} f = 2}$$

2) simplification: $E_2, E_{21}, E_{11}, E_{22}, E_{11}+E_{22} = \text{linearly indep. in } M_2(\mathbb{R})$;

det. $M_f = ?$

$$\frac{E_{12} - E_{21}}{2} = E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \frac{E_{21}}{2} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \frac{E_{11} - E_{22}}{2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix};$$

$$A_4 = E_{11} + E_{22} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Studium lin. ind. matrices A_1, A_2, A_3, A_4 . Für $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{R}, a \neq 0$:

$$\lambda A_1 + \lambda_2 A_2 + \lambda_3 A_3 = 0_2 \Rightarrow \begin{cases} \lambda_3 + \lambda_4 = 0 \\ \lambda_1 = 0 \\ \lambda_2 = 0 \\ \lambda_4 - \lambda_3 = 0 \Rightarrow \lambda_4 = \lambda_3 = 0 \end{cases} \Rightarrow 2\lambda_3 = 0 \Rightarrow \lambda_3 = 0 \Rightarrow \lambda_4 = 0,$$

$\Rightarrow A_1, A_2, A_3, A_4 = \text{lin. ind.} \Rightarrow B = \text{Sp}\{A_1, A_2, A_3, A_4\} = \text{linear}$

$$M^B(f) = ?$$

$$f(E_{12}) = A \cdot E_{12} - E_{12} \cdot A = - \begin{pmatrix} 6 & 1 \\ 8 & 4 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 6 & 1 \\ 8 & 4 \end{pmatrix} \\ = - \begin{pmatrix} 0 & 6 \\ 0 & 8 \end{pmatrix} + \begin{pmatrix} 8 & 4 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 8 & -2 \\ 0 & -8 \end{pmatrix}$$

$$c) \text{Ker}(f) \oplus \text{Im} f \stackrel{?}{=} \mathcal{M}_2(\mathbb{R})?$$

$$\text{Ker}(f) = \text{Sp} \left\{ \begin{pmatrix} 2 & 1 \\ 8 & 0 \end{pmatrix}; \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$$\text{Im}(f) = \text{Sp} \left\{ \begin{pmatrix} 1 & 0 \\ -10 & -1 \end{pmatrix}; \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\}$$

$$\text{Im}(f) + \text{Ker} f = \text{Sp} \left\{ \underbrace{\begin{pmatrix} 2 & 1 \\ 8 & 0 \end{pmatrix}}_{B_1}; \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{B_2}; \underbrace{\begin{pmatrix} 1 & 0 \\ -10 & -1 \end{pmatrix}}_{B_3}; \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{B_4} \right\}$$

Studium lin. ind. lui B_1, B_2, B_3, B_4 : Fie $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$ a. n.

$$\alpha_1 B_1 + \alpha_2 B_2 + \alpha_3 B_3 + \alpha_4 B_4 = \mathbf{0}_2 \Rightarrow \begin{cases} 2\alpha_1 + \alpha_2 + \alpha_3 = 0 \\ \alpha_1 + \alpha_4 = 0 \Rightarrow \alpha_4 = -\alpha_1 \\ 8\alpha_1 - 10\alpha_3 = 0 \Rightarrow \alpha_1 = \frac{5}{4}\alpha_3 \\ \alpha_2 - \alpha_3 = 0 \Rightarrow \alpha_2 = \alpha_3 \end{cases}$$

$$2\alpha_1 + \alpha_2 + \alpha_3 = 0 \Rightarrow -\frac{5}{2}\alpha_3 + \alpha_3 + \alpha_3 = 0 \Rightarrow \alpha_3 = 0 \Rightarrow \alpha_1 = \alpha_2 = \alpha_4 = 0 \Rightarrow$$

$$\Rightarrow B_1, B_2, B_3, B_4 = \text{lin. ind.} \Rightarrow \text{Ker}(f) \cap \text{Im} f = \{\mathbf{0}_2\} \Rightarrow \text{Ker}(f) \oplus \text{Im}(f) =$$

$$= \text{suma directa} \Rightarrow \text{zi deoarece } B_1, B_2, B_3, B_4 = \text{lin. ind.} \Rightarrow$$

$$\Rightarrow \text{Ker}(f) \oplus \text{Im}(f) = \mathcal{M}_2(\mathbb{R})$$

$$4) U = \text{Sp} \left\{ u_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}; u_2 = \begin{pmatrix} -3 \\ 6 \\ 0 \end{pmatrix}; u_3 = \begin{pmatrix} -5 \\ 14 \\ 2 \end{pmatrix} \right\}, \quad u = \begin{pmatrix} 12 \\ 0 \\ 12 \end{pmatrix}$$

$$a) \cos(\angle(u_1, u_2)) = \frac{\langle u_1, u_2 \rangle}{\|u_1\| \|u_2\|} = \frac{-6 - 12}{\sqrt{9} \cdot \sqrt{9+36}} = \frac{-18}{3\sqrt{45}} = -\frac{18}{3 \cdot 3\sqrt{5}} = -\frac{2}{\sqrt{5}} \quad (=) \quad \boxed{\cos(\angle(u_1, u_2)) = -\frac{2}{\sqrt{5}}}$$

$$b) \langle u_1, u_2 \rangle = -18 \Rightarrow u_1 \perp u_2$$

Aplicăm algoritmul Gram-Schmidt:

$$u_1 = u_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$u_2 = u_2 - \text{pr}_{u_1} u_2 = u_2 - \frac{\langle u_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = u_2 - \frac{-18}{9} \cdot u_1 =$$

$$= u_2 + 2u_1 = u_2 + 2u_1 = \begin{pmatrix} -3 \\ 6 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad (=) \quad u_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$u_3 = u_3 - \text{pr}_{u_1} u_3 - \text{pr}_{u_2} u_3 = u_3 - \frac{\langle u_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle u_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 =$$

$$= u_3 - \frac{-36}{9} \cdot u_1 - \frac{24}{9} \cdot u_2 = u_3 + 4u_1 - 3u_2 = \begin{pmatrix} -5 \\ 14 \\ 2 \end{pmatrix} + \begin{pmatrix} 8 \\ -8 \\ 4 \end{pmatrix} -$$

$$- \begin{pmatrix} 3 \\ 6 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 6 \\ 6 \end{pmatrix} = 0$$

Completăm la o bază ortogonală astfel: căutăm $u_3 \perp u_1$ și $u_3 \perp u_2 \Rightarrow$

$$(\Rightarrow) \begin{cases} \langle u_3, u_1 \rangle = 0 \\ \langle u_3, u_2 \rangle = 0 \end{cases}$$

$$u_3 = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \langle u_3, u_1 \rangle = 0 \Rightarrow 2a - 2b + c = 0$$

$$\langle u_3, u_2 \rangle = 0 \Rightarrow a + 2b + 2c = 0 +$$

$$a + c = 0 \Rightarrow c = -a \Rightarrow 2b = -a - 2c = -a + 2a = a,$$

$$\Rightarrow b = \frac{a}{2}$$

$$u_3 = \begin{pmatrix} a \\ \frac{a}{2} \\ -a \end{pmatrix}$$

$$\text{Put } a=2 \Rightarrow u_3 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Forman o least orthonormal:

$$q_1 = \frac{1}{\|u_1\|} \cdot u_1 = \frac{1}{3} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$q_2 = \frac{1}{\|u_2\|} \cdot u_2 = \frac{1}{3} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$q_3 = \frac{1}{\|u_3\|} \cdot u_3 = \frac{1}{3} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

$\{q_1, q_2, q_3\}$ = least orthonormal

$$\begin{aligned} pr_u &= pr_{q_1} + pr_{q_2} + pr_{q_3} = \frac{\langle u, q_1 \rangle}{\langle q_1, q_1 \rangle} \cdot q_1 + \frac{\langle u, q_2 \rangle}{\langle q_2, q_2 \rangle} \cdot q_2 + \\ &+ \frac{\langle u, q_3 \rangle}{\langle q_3, q_3 \rangle} \cdot q_3 = \frac{1}{3} (24 + 12) \cdot q_1 + \frac{1}{3} (12 + 24) \cdot q_2 + \frac{1}{3} (24 - 24) \cdot q_3 \end{aligned}$$

$$= \frac{4q_1 + 4q_2}{3} = \frac{12(q_1 + q_2)}{3} = 4 \cdot \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 16 \\ 0 \\ 12 \end{pmatrix} \quad (=1)$$

$$\Rightarrow pr_u = \begin{pmatrix} 16 \\ 0 \\ 12 \end{pmatrix}$$

$$c) \sum_{i=1}^3 \|u - q_i\|^2 \geq 3 \|u\|^2 - 2\sqrt{3} \|u\| + 3$$

$$\text{Hence } |\langle u, u \rangle| \leq \|u\| \cdot \|u\|, \quad u, u \in \mathbb{R}^3$$

$$\|u - q_i\|^2 \geq \|u - q_i\| \cdot \|u - q_i\| \geq \langle u - q_i, u - q_i \rangle =$$

$$= \langle u, u - q_i \rangle - \langle q_i, u - q_i \rangle = \langle u, u \rangle - \langle u, q_i \rangle - \langle q_i, u \rangle + \langle q_i, q_i \rangle =$$

$$= \left\| u \right\|^2 - 2 \langle u, z_i \rangle + \left\| z_i \right\|^2 = \left\| u \right\|^2 - 2 \langle u, z_i \rangle + 1$$

Insammeln:

$$\sum_{i=1}^3 \left\| u - z_i \right\|^2 \geq 3 \left\| u \right\|^2 - 2 \sum_{i=1}^3 \langle u, z_i \rangle + 3$$

$$u = \begin{pmatrix} a \\ a \\ c \end{pmatrix}$$

$$z_1 = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \Rightarrow \langle u, z_1 \rangle \leq \left\| u \right\| \cdot \frac{1}{\sqrt{3}} \Rightarrow -\langle u, z_1 \rangle \geq -\frac{1}{\sqrt{3}} \left\| u \right\|$$

$$\text{Dci} \quad \sum_{i=1}^3 \left\| u - z_i \right\|^2 \geq 3 \left\| u \right\|^2 - 2\sqrt{3} \cdot \left\| u \right\| + 3$$

$$3) a) \quad A = \begin{matrix} & \begin{matrix} AP & S & AN \end{matrix} \\ \begin{matrix} AP \\ S \\ AN \end{matrix} & \begin{pmatrix} 0,75 & 0,25 & 0,20 \\ 0,05 & 0,25 & 0,55 \\ 0,2 & 0,5 & 0,25 \end{pmatrix} \end{matrix}$$

$$\begin{matrix} 0,75 \\ AP \end{matrix} \xrightarrow{0,25} S \xrightarrow{0,25} AN$$

$$c) \quad u(m) = A \cdot u(m) \quad \text{et}$$

$$m=0 \Rightarrow u(1) = A \cdot u(0)$$

$$m=1 \Rightarrow u(2) = A \cdot u(1) = A^2 \cdot u(0)$$

Inductive ansatz: $u(m) = A^m \cdot (u_0)$, mit A determiniert (a)

$$u(2) = A^2 \cdot u(0), \quad u(0) = \begin{pmatrix} 1 \\ 0,25 \\ 0,2 \end{pmatrix} \Rightarrow u(2) = \begin{pmatrix} 0,75 & 0,25 & 0,2 \\ 0,05 & 0,25 & 0,55 \\ 0,2 & 0,5 & 0,25 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0,25 \\ 0,2 \end{pmatrix} = \begin{pmatrix} 0,75 \cdot 1 + 0,25 \cdot 0,25 + 0,2 \cdot 0,2 \\ 0,05 \cdot 1 + 0,25 \cdot 0,25 + 0,55 \cdot 0,2 \\ 0,2 \cdot 1 + 0,5 \cdot 0,25 + 0,25 \cdot 0,2 \end{pmatrix}$$

$$p_A(x) = \begin{vmatrix} 0,75-x & 0,25 & 0,20 \\ 0,05 & 0,25-x & 0,55 \\ 0,2 & 0,5 & 0,25-x \end{vmatrix} = 0 \Leftrightarrow \begin{cases} \lambda_1 = \dots \\ \lambda_2 = \dots \\ \lambda_3 = \dots \end{cases}$$

Also $v_{\lambda_1}, v_{\lambda_2}, v_{\lambda_3}$ und $v_\lambda = \{ \lambda \in \mathbb{R}^3 / |\lambda| = 1 \}$

Formen $p = (u_1, u_2, u_3)$, und $u_1, u_2, u_3 \in v_{\lambda_1}, v_{\lambda_2}, v_{\lambda_3}$

$$A = p \cdot D \cdot p^{-1}, \text{ mit } D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \Rightarrow A^n = p \cdot D^n \cdot p^{-1} \Rightarrow$$

$$\Rightarrow u(30) = A^{30} \cdot u(0) = p D^{30} \cdot p^{-1} \cdot u(0), \text{ mit } u(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$